

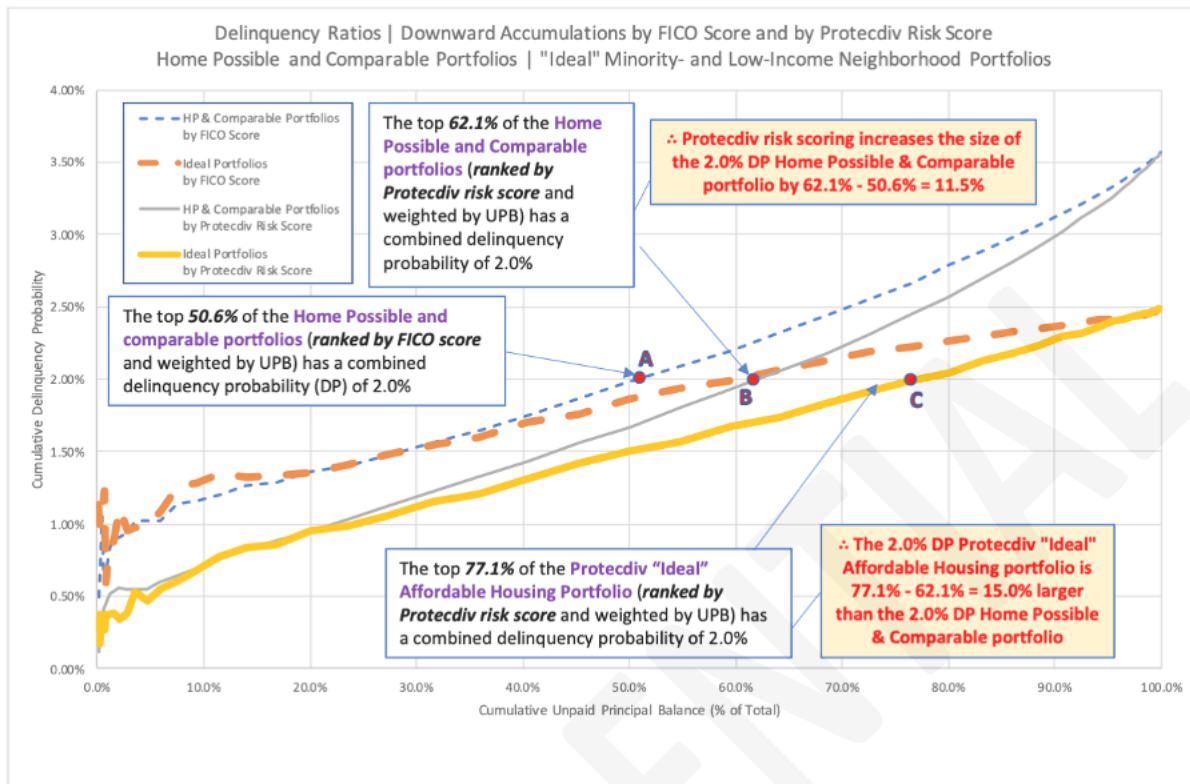
The global reinsurance marketplace has a strong opportunity to bring reinsurance capacity to the low income and affordable housing mortgage markets. Analyses of the recent performance of loans in the affordable housing mortgage markets prove that reinsurers have a reason to be excited, rather than afraid, of the risk in this market segment.

Capital providers are already underwriting reinsurance for affordable housing mortgages (for example, through Freddie Mac's Home Possible program) at low but positive margins, with reasonable volatility.ⁱ This is proof that reinsurers can accomplish their Environmental, Social, and Governance (ESG) goals profitably – and there's room for improvement. Using analytics, we can look beyond the usual mortgage reinsurance underwriting criteria to define a broader set of risk selection variables and more accurately identify mortgage default risks. This can help increase margins, reduce volatility, and define a more optimal universe of mortgage reinsurance loan portfolios.

Society has an appetite to see access to homeownership improve for underserved ethnic-minority and low-income communities. Even this year, a top-five American bank was found to be systematically avoiding the issuance of mortgages in minority and low-income neighborhoods, under the assumption that these neighborhoods bore inherently high mortgage default risk. We contend that there are thousands of homeowners in these neighborhoods who have demonstrably low default risk, and what we're proposing is that we can use more detailed and thoughtful analytics to progress toward four socially beneficial outcomes:

- Minimizing the blanket perception of affordable housing mortgages as high risks.
- Improving the availability of mortgages to financially stable and responsible homeowners in low-income neighborhoods.
- Reducing mortgage reinsurance costs.
- Reducing the cost of homeownership overall.

Analytics can't solve all the problems of affordable housing mortgage availability, but they can be used to show that the costs of affordable-housing lending need not be feared.



Consider a portfolio of mortgages from Freddie Mac's Home Possible affordable housing homeownership program and a similarly sized portfolio of comparable mortgages outside the Home Possible program. The mortgage insurance specialists from the actuarial consultancy Milliman have collaborated with Protecdiv to gather all publicly available origination and development data for the Home Possible program; in addition, they have applied leading data science construct a comparable non-Home Possible portfolio.ⁱⁱ Together, the Home Possible and comparable non-Home Possible portfolios have a delinquency probability of slightly more than 3.5%. One way to rank the mortgages in these portfolios is in descending order by FICO credit score. Conceivably, a government-sponsored entity (GSE) such as Freddie Mac could construct a new mortgage pool from these mortgages by accumulating them in rank order, i.e., in descending order by FICO score: put the mortgage with the highest FICO score into the empty portfolio first, followed by the mortgage with the second-highest FICO score, then the one with third-highest FICO score, and so on. If such a **NEW MORTGAGE POOL** were built up this way, the top 50.6% of the mortgages (weighted by unpaid principal balance) would have a combined delinquency probability of about 2.0%.ⁱⁱⁱ

Is there a better way to accumulate the portfolio? In other words, is there another way to rank the mortgages so that a higher percentage of the Home Possible and comparable mortgages' UPB combines for a 2.0% delinquency probability? Protecdiv's study says yes, there is, if we score each mortgage's risk based on a weighted average of its percentile rank in four variables:

- 45.9% of the mortgage's FICO score percentile rank
- + 22.4% of the mortgage's debt-to-income (DTI) ratio percentile rank
- + 16.4% of the mortgage's combined loan-to-value (CLTV) ratio percentile rank
- + 15.3% of the mortgage's tropical storm vulnerability percentile rank

This weighting produces a risk score between 0 (for the lowest FICO score, highest DTI ratio, highest CLTV ratio, and most vulnerable coastal tropical storm vulnerability) to 100 (for the highest FICO score, lowest DTI ratio, lowest CLTV ratio, and least vulnerable non-coastal tropical storm vulnerability). When the **NEW MORTGAGE POOL** of Home Possible and comparable mortgages is constructed based on this score, 62.1% of the mortgages combine for a 2.0% delinquency probability. This is an improvement of $62.1\% - 50.6\% = 11.5\%$. In other words, by replacing FICO score with the weighted average risk score, the GSE can put 11.5% more UPB into the **NEW MORTGAGE POOL** without making the delinquency probability any worse.

The Home Possible portfolio contains mortgages from metropolitan statistical areas (MSAs) that have low percentage minority populations and high average household incomes. If the GSE were to want to build up a new and more intentionally affordable-housing-focused portfolio, it would be helpful to require that the mortgage come from an MSA with either a high percentage minority population or a low average household income (one or the other, or both). Can the GSE do this without making the delinquency probability any worse? The answer is yes, it can, if the portfolio is restricted to mortgages with weighted average percentile risk scores of at least 40 – in other words, if the portfolio is made up of the best 70% of the UPB volume of the Home Possible and comparable portfolios (as 70% of the portfolio has scores of at least 40). When the **NEW MORTGAGE POOL** is built up under these constraints – a weighted average percentile risk score of 40 or more, in an MSA with either a minority population of at least 63% or an average household income no greater than \$48,000 – then 77.1% of the portfolio will combine for a 2.0% delinquency probability. This would be a $77.1\% - 62.1\% = 15.0\%$ improvement over the unconstrained (all risk scores, all minority population percentages, all income levels) portfolio.

In summary:

Portfolio	Subportfolio			Ranking Basis	UPB, Best-Ranked Risks (\$MM)	UPB, All Risks (\$MM)	% of Best-Ranked Risks Aggregating to a 2.0% Default Probability
	Protecdiv Risk Scores	% Minority Population in MSA	Average Income in MSA				
Home Possible + Comparable Loans	0-100	and (any or any)		FICO Score	\$ 105,866 MM	\$ 209,391 MM	50.6%
Home Possible + Comparable Loans	0-100	and (any or any)		Protecdiv Risk Score	\$ 130,111 MM	\$ 209,391 MM	62.1%
Home Possible + Comparable Loans	40-100	and (63.0%+ or \$0 - \$48,000)		Protecdiv Risk Score	\$ 25,169 MM	\$ 32,653 MM	77.1%

Thus, with a combination of a new risk scoring methodology and the application of that methodology in risk selection, the universe of desirable mortgage reinsurance risks can be grown by a large multiple (in this case, growth by a factor of $77.1\% / 50.6\% = 1.52$) without worsening the portfolio's delinquency probability. This is an improvement in risk that promises to invite reinsurance capital from several new sources.^{iv}

ⁱ In particular, the Federal Housing Administration (FHA) has been providing affordable-housing mortgage reinsurance for years and has maintained a strong capital ratio over time.

ⁱⁱ Milliman's comparable portfolio mirrors the Home Possible portfolio in origination years, borrower FICO scores, borrower debt-to-income ratios, and mortgage loan-to-value ratios. The "comparable portfolio" has roughly the same volume of unpaid principal balances (UPB) as the universe of Freddie Mac's Home Possible mortgages, and therefore roughly doubles the volume of data available to Protecdiv for analysis. Protecdiv is deeply grateful to Milliman for their analytic services and market insights, many of which underpin the analysis presented throughout this whitepaper.

ⁱⁱⁱ A 2.0% delinquency probability correlates roughly to the 15 basis points of default risk typically underwritten by mortgage reinsurance underwriters:

	a	b	c	d	e = acd
	Mortgage Delinquency Rate				
Housing Market Conditions	Annual Probability	Multiple of Normal	Rate under Given Conditions	Expected Loss Severity per Delinquency	Expected Annual Severity
Normal	90%	1.00x	2.0%	0.00%	0.00%
Stressed	10%	2.50x	5.0%	30.00%	0.15%
All	100%	1.15x	2.3%	6.52%	0.15%

^{iv} The reinsurance market's memory of the 2007-2010 mortgage crisis raises concerns as to whether borrowers will be able to repay their mortgages if a crisis of a similar magnitude arises and fuels a new wave of delinquencies and strains on repayment. The counterargument to these concerns is the observation that the low-income borrowers who meet Home Possible's current underwriting criteria are rigorously educated in the best practices of financial responsibility and home care. Today's low-income borrowers are less susceptible to payment stress surprises than their counterparts from the early to mid-Aughts, and in many cases are even less susceptible than present-day borrowers outside the Home Possible program.